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Transformer filter with notch

Abstract

A device includes a filter circuit having both a transformer and a notch filter. The notch filter is formed via capacitive cross-coupling of windings of the transformer. The transformer includes a first winding with an input terminal and an output terminal and a second winding with an input terminal and an output terminal. The notch filter is formed by coupling a first capacitor between the input terminal of the first winding and the output terminal of the second winding, and by coupling a second capacitor between the output terminal of the first winding and the input terminal of the second winding.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority under 35 U.S.C. § 119 to Indian Patent Application number 202341067079, filed Oct. 6, 2023, the contents of which are incorporated by reference herein.

BACKGROUND

(2) Many electronic devices employ a digital-to-analog converter (DAC) to generate analog signals for various applications. For example, some radar systems synthesize a radar signal in the digital domain and employ a DAC to convert the digital signal to an analog signal for transmission. However, in at least some cases the DAC introduces unwanted components (e.g., images and harmonics) into the analog signal. Some devices employ a filter, such as a tank filter, to reduce the impact of the unwanted components on the analog signal. However, conventional filters have a relatively limited impact on the unwanted components and are thus not well-suited for some devices and applications, such as radar applications wherein the analog output signal of the DAC is fed to a multiplier.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.

(2) FIG. 1 is a block diagram of a filter circuit that employs a transformer and a notch filter in accordance with some embodiments.

(3) FIG. 2 is a diagram illustrating an example behavior of the filter circuit of FIG. 1 in accordance with some embodiments.

(4) FIG. 3 is a block diagram of a device including a DAC and the filter circuit of FIG. 1 in accordance with some embodiments.

(5) FIG. 4 is a diagram illustrating routing of portions of the notch filter of FIG. 1 in accordance with some embodiments.

DETAILED DESCRIPTION

(6) FIGS. 1-4 illustrate circuits and devices including a filter circuit having both a transformer and a notch filter in accordance with some embodiments. The notch filter is formed via capacitive cross-coupling of windings of the transformer. For example, in some embodiments the transformer includes a first winding with an input terminal and an output terminal and a second winding with an input terminal and an output terminal. The notch filter is formed by connecting a first capacitor between the input terminal of the first winding and the output terminal of the second winding, and by connecting a second capacitor between the output terminal of the first winding and the input terminal of the second winding. This configuration results in the frequency response of the overall filter circuit having a notch at a particular frequency, wherein the notch frequency is based on the capacitive value of the capacitors used in the cross-coupling. This notch, together with the filter characteristics of the transformer itself, allow the overall filter circuit to provide increased rejection of unwanted signal components in an input signal, without adding much noise or non-linearity to the output signal. The filter circuit is thus well-suited for applications that require relatively high rejection of images and harmonics.

(7) To illustrate via an example, some radar devices, such as Frequency-Modulated Continuous Wave (FMCW) radar systems, synthesize a radar signal in the digital domain, then employ a DAC to convert the digital radar signal to an analog signal, multiply the analog signal to increase the signal amplitude for transmission, and then transmit the multiplied signal. However, the DAC can introduce unwanted images and harmonics into the analog signal, and the multiplication of the signal can exacerbate the impact of the unwanted images and harmonics, and thus degrade the overall performance of the radar device. Conventionally, a filter is employed to reject the unwanted images and harmonics from the analog output signal of the DAC. However, conventional filters typically do not provide sufficiently high signal rejection (e.g., providing 38 dB rejection or less). Furthermore, these conventional filters can add noise and distortion to the analog signal.

(8) In contrast to these conventional filters, described herein are embodiments of a filter circuit that employs two components: a transformer and a notch filter. The transformer provides magnetic coupling of an input signal and an output signal that gives two resonating peaks, wherein the separation between the peaks depends on the level of the magnetic coupling. The effect of these resonating peaks is to provide a first order slope in the frequency response of the transformer at lower frequencies and a third order slope in the frequency response of the transformer at higher frequencies. This results in a relatively high signal rejection (e.g., 44 dB).

(9) The notch filter is formed by capacitive cross-coupling the windings of the transformer. This creates high levels of signal rejection, sometimes referred to as “zeroes”, at and around a notch frequency. The notch frequency is based at least in part on the level of capacitance (that is, the size of the capacitors) employed in the capacitive cross-coupling. Thus, by selecting and employing capacitors of a particular value, the notch frequency of the notch filter is set to a corresponding level. Accordingly, in some embodiments the capacitors are set to a value such that the filter circuit has increased rejection at and around the frequency of the unwanted images and harmonics, thus improving the overall performance of the filter circuit and the corresponding radar system.

(10) FIG. 1 illustrates a filter **100** in accordance with some embodiments. In the illustrated example, the filter **100** includes a current source **102**, a current sink **103**, a transformer **104** including a first (primary) winding **105** and a second (secondary) winding **106**, and further includes capacitors **108**, **109**, **115**, and **116**. The current source **102** includes a first terminal connected to a ground voltage reference. The current sink **103** includes a first terminal connected to the ground voltage reference and a second terminal connected to a node **112**. The capacitor **115** includes a first terminal connected to the node **110** and a second terminal connected to the node **112**.

(11) The first winding **105** includes an input terminal connected to the node **110** and an output terminal connected to the node **112**. The second winding **106** includes an input terminal connected to a node **113**, and an input terminal connected to a node **111**. The capacitor **116** includes a first terminal connected to the node **111** and a second terminal connected to the node **113**. The capacitor **108** includes a first terminal connected to the node **110** and a second terminal connected to the node **111**. The capacitor **109** includes a first terminal connected to the node **112** and a second terminal connected to the node **113**.

(12) In operation, the current source **102** applies a current at node **110**, thus charging the capacitor **115** to a voltage based on the amplitude of the current. The capacitor discharges to the ground reference via the current sink **103**. In at least some embodiments, the amplitude of the current supplied by the current source **102** changes over time, such that the supplied current, and therefore the charge at the capacitor **115**, and the voltage across the nodes **110** and **112**, represent an input signal for the filter **100**.

(13) The transformer **104** transforms the input voltage across the nodes **110** and **112** to an output voltage across the nodes **111** and **113**, and therefore across the capacitor **116**. The relationship between the output voltage and input voltage is governed by the ratio in the number of coil turns between the winding **105** and the winding **106**. That is, the ratio of the input voltage to the output voltage is equal to the ratio of the number of turns in the winding **105** to the number of turns in the

winding **106**. In addition, when an alternating current signal is applied at the input to the transformer **104**, the windings **105** and **106** act as two resonators that couple magnetically, thus providing voltage transformation that varies according to the frequency of the input signal. In particular, the magnetic coupling applies two resonating peaks such that the frequency response of the transformer **104** has a first order slope at lower frequencies and a third order slope at higher frequencies. This third order slope allows the transformer **104** to be used as a filter that rejects higher frequencies, wherein the rejected frequencies are based on coupling factor, designated K, between the windings **105** and **106**.

(14) The capacitors **108** and **109** are connected such that the windings **105** and **106** are capacitively cross-coupled. That is, the capacitor **108** cross-couples the input of the winding **105** to the output of the winding **106**. Further, the capacitor **109** cross-couples the input of the winding **106** to the output of the winding **105**. This capacitive cross coupling changes the overall frequency response of the filter **100**, and in particular places zeroes in the frequency response, thus creating a notch in the frequency response at a notch frequency. The zeroes in the frequency response are set according to the following formula:

$$(15) \frac{1}{2} \sqrt{\frac{2K_m}{L(1 - K_m^2)C_c}}$$

where L is the inductance of the windings **105** and **106**, Km is the magnetic coupling coefficient for the transformer **104**, and Cc is the capacitive values for the capacitors **108** and **109**. In some embodiments, the capacitors **108** and **108** have substantially equal (e.g., within one percent) capacitive values.

(16) Thus, as shown by the formula above, the placement of the zeroes, and the corresponding notch, can be set by selecting the capacitors **108** and **109** of a particular corresponding value. This allows the filter **100** to have increased signal rejection around a desired frequency (as set by the capacitors **108** and **109**), thus improving the performance of the filter **100**. In addition, the filter **100** uses only passive devices (e.g., the transformer **104** and the capacitors **108** and **109**), and therefore adds relatively little noise and non-linearity to the input signal.

(17) FIG. 2 illustrates a diagram **200** depicting an example frequency response for the filter **100** in accordance with some embodiments. The diagram **200** includes an x-axis representing frequency (in Hertz) and a y-axis representing voltage attenuation (also referred to as rejection) expressed in negative decibels, or dB. The diagram **200** also includes a curve **220**, representing the frequency response for the filter **100**. In particular the curve **220** indicates, for each frequency, the corresponding amount of rejection for a signal, or signal component, at that frequency.

(18) In the illustrated example, the curve **220** has a peak **221** at or about 2.5 GHz. This is referred to as the center frequency for the filter **100**. As shown, the curve **220** has a substantially first order slope prior to the center frequency and a substantially third order slope at frequencies higher than the center frequency. These different slopes, and thus the different levels of frequency rejection for lower and higher frequencies, are a result of the magnetic coupling between the windings **105** and **106** creating two resonating peaks. The separation between the peaks depends on the level of magnetic coupling between the windings **105** and **106**. Further, the third order slope of the curve **220** for higher frequencies indicates a relatively high level of rejection for these higher frequencies compared to at least some conventional tank filters.

(19) In addition, the curve **220** has a notch **222**, representing a relatively high level of rejection at a corresponding notch frequency. This notch is created by the capacitive cross-coupling between the windings **105** and **106** of the transformer **104**. The notch frequency is based on the capacitive values of the capacitors **108** and **109**, as shown above. Thus, by selecting the appropriate capacitor values, the notch frequency for the filter **100** can be selected. This allows the filter **100** to be configured to have increased rejection around a particular frequency of interest. For example, in some embodiments, the notch frequency is set (by selecting the corresponding capacitive values for the capacitors **108** and **109**) such that the notch frequency is at the frequency of an unwanted image

or harmonic of an analog input signal provided to the filter **100**.

(20) FIG. 3 illustrates a diagram of aspects of a device **300** that incorporates the filter **100** in accordance with some embodiments. In the illustrated example, the device **300** includes the filter **100**, a radiofrequency (RF) DAC **325**, and a multiplier **326**. The RF DAC **325** is configured to receive a digital input code and generate an output signal (e.g., a sinusoidal output signal) having a frequency based on the digital input code, and with a sampling clock of frequency $f_{\text{sub.s}}$. The DAC **325** can be any type of DAC configured to generate an analog output signal having a frequency based on a digital input code (e.g., a delta-sigma DAC, resistor-based DAC, switch-based DAC, and the like).

(21) The filter **100** is connected to the RF DAC **325** to receive the analog output signal, to filter the received analog signal via the transformer **104** and the capacitive cross-coupling of the capacitors **108** and **109**, as explained above. The filter **100** thus generates a filter output signal, representing a filtered representation of the analog signal generated by the DAC **325**. The multiplier **326** is circuitry configured to receive the filter output signal and multiply the amplitude of that signal by a specified amount, thus generating a multiplied output signal. The multiplied output signal is provided to other circuitry of the device **300** (not shown) for further processing, such as one or more of mixing, amplification, filtering, and transmission.

(22) By employing the filter **100** between the RF DAC **325** and the multiplier **326**, the device **300** is able to generate a multiplier output signal that is relatively free of unwanted images and harmonics, collectively referred to as a “frequency spur”, that is generated by the RF DAC **325**. To illustrate via an example, in some embodiments the device **300** is part of an FMCW radar system, and in particular is part of a signal generator configured to generate a chirp signal for a radar front end of the FMCW radar system. The analog output signal generated by the RF DAC **325** includes a frequency spur that includes unwanted images (that is, unwanted signal components) at or around frequencies that are multiples of the frequency indicated by the input digital code. Without the filter **100**, the frequency spur of the DAC output signal would be multiplied by the multiplier **326** at their generated amplitudes. This would result in the chirp signal of the FMCW radar system having high levels of unwanted images and harmonics, resulting in poor system performance. By including the filter **100**, the device **300** is able to substantially reject the frequency spur. In particular, the values of the capacitors **108** and **109** are selected such that the corresponding notch frequency of the filter **100** is at or around the frequency of at least one of the unwanted images, resulting in high rejection of the unwanted image. This results in a chirp signal that is relatively free of the unwanted images and harmonics, improving overall system performance.

(23) In some cases, the routing of the connections for the cross-coupled capacitors **108** and **109** results in parasitic resistance and parasitic inductance at the filter **100**. This parasitic resistance and inductance can degrade both the notch (that is, reduces the amount of voltage rejection associated with the notch) and the amount of rejection at higher frequencies, thus degrading the overall performance of the filter **100**. Accordingly, in some embodiments the connections of the capacitors **108** and **109** are routed to reduce the amount of parasitic inductance and resistance and thus improve overall filter performance. An example is illustrated at FIG. 4 in accordance with some embodiments.

(24) In the illustrated example, the filter **100** is part of an integrated circuit device **440**. The integrated circuit device **440** is or includes a semiconductor device having at least two semiconductor layers, designated layers **441** and **442**, wherein layer **442** is formed such that it is located underneath layer **441**. Furthermore, in the depicted example, the capacitors **108** and **109** are connected to the windings **105** and **106** of the transformer **104** via a set of interconnects. In particular, the capacitor **108** has a first terminal connected to the input of the winding **105** via an interconnect **446** and a second terminal connected to the output of the winding **106** via an interconnect **448**. The capacitor **109** has a first terminal connected to the input of the winding **106** via an interconnect **447** and a second terminal connected to the output of the winding **105** via an

interconnect **445**.

(25) The windings **105** and **106**, as well as the capacitors **108** and **109**, are formed such that these components are located at the semiconductor layer **441** of the integrated circuit **440**. However, the interconnects **445**, **446**, **447**, and **448** are formed such that a portion of each interconnect is located at the semiconductor layer **442**. Thus, the cross-coupled connections of the capacitors **108** and **109** to the windings **105** and **106** are routed underneath the components of the filter **100**. In at least some cases, this routing shortens the overall length of at least one of the interconnects **445**, **446**, **447**, and **448**. This reduced length reduces the parasitic resistance and inductance associated with the filter **100**, and thus improves the overall filter performance.

(26) In some embodiments, certain aspects of the techniques described above may be implemented by one or more processors of a processing system executing software. The software comprises one or more sets of executable instructions stored or otherwise tangibly embodied on a non-transitory computer readable storage medium. The software can include the instructions and certain data that, when executed by the one or more processors, manipulate the one or more processors to perform one or more aspects of the techniques described above. The non-transitory computer readable storage medium can include, for example, a magnetic or optical disk storage device, solid state storage devices such as Flash memory, a cache, random access memory (RAM) or other non-volatile memory device or devices, and the like. The executable instructions stored on the non-transitory computer readable storage medium may be in source code, assembly language code, object code, or other instruction format that is interpreted or otherwise executable by one or more processors.

(27) A computer readable storage medium may include any storage medium, or combination of storage media, accessible by a computer system during use to provide instructions and/or data to the computer system. Such storage media can include, but is not limited to, optical media (e.g., compact disc (CD), digital versatile disc (DVD), Blu-Ray disc), magnetic media (e.g., floppy disc, magnetic tape, or magnetic hard drive), volatile memory (e.g., random access memory (RAM) or cache), non-volatile memory (e.g., read-only memory (ROM) or Flash memory), or microelectromechanical systems (MEMS)-based storage media. The computer readable storage medium may be embedded in the computing system (e.g., system RAM or ROM), fixedly attached to the computing system (e.g., a magnetic hard drive), removably attached to the computing system (e.g., an optical disc or Universal Serial Bus (USB)-based Flash memory) or coupled to the computer system via a wired or wireless network (e.g., network accessible storage (NAS)).

(28) Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed is not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

(29) Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular embodiments disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore

evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A device comprising: a digital-to-analog converter (DAC) having at least one output; a transformer including a first winding and a second winding, wherein the at least one output of the DAC is directly connected to the first winding of the transformer; and a notch filter including a first capacitive cross-coupling of the first winding and the second winding, wherein the DAC is configured to provide an output signal to the transformer, the output signal having a frequency spur, and the notch filter is configured to substantially reject the frequency spur.
2. The device of claim 1, wherein: the notch filter comprises a second capacitive cross-coupling of the first winding and the second winding.
3. The device of claim 2, wherein: the first capacitive cross-coupling includes a first capacitor having a first terminal connected to an input terminal of the first winding and a second terminal connected to an output terminal of the second winding.
4. The device of claim 3, wherein: the second capacitive cross-coupling includes a second capacitor having a first terminal connected to an output terminal of the first winding and a second terminal connected to an input terminal of the second winding.
5. The device of claim 4, wherein: the first capacitor and the second capacitor have substantially equal capacitances.
6. The device of claim 1, wherein: the device includes a semiconductor having a first layer and a second layer; the first winding and the second winding are located at the first layer of the semiconductor; and at least a portion of the first capacitive cross-coupling is located at the second layer of the semiconductor.
7. The device of claim 6, wherein the second layer is located under the first layer.
8. The device of claim 1, further comprising: a multiplier coupled to an output of the transformer.
9. A semiconductor device comprising: a digital-to-analog converter (DAC) including an output to provide an output signal; a transformer connected directly to the output of the DAC; a notch filter coupled to the transformer, the notch filter including: a first capacitive cross-coupling of a first winding the transformer and a second winding of the transformer; and a first layer and a second layer, wherein the first winding and the second winding are located at the first layer, and at least a portion of the first capacitive cross-coupling is routed via the second layer.
10. The semiconductor device of claim 9, wherein: the notch filter comprises a second capacitive cross-coupling of the first winding and the second winding.
11. The semiconductor device of claim 10, wherein: the first capacitive cross-coupling includes a first capacitor having a first terminal connected to an input terminal of the first winding and a second terminal connected to an output terminal of the second winding.
12. The semiconductor device of claim 11, wherein: the second capacitive cross-coupling includes a second capacitor having a first terminal connected to an output terminal of the first winding and a second terminal connected to an input terminal of the second winding.
13. The semiconductor device of claim 12, wherein: the first capacitor and the second capacitor have substantially equal capacitances.
14. The device of claim 9, wherein the second layer is located under the first layer.
15. A method, comprising: providing, by a digital-to-analog converter (DAC) that is directly connected to a transformer, an input signal directly to the transformer; filtering the input signal at the transformer, the transformer including a first winding and a second winding; notch filtering the input signal at a first capacitive cross-coupling of the first winding and the second winding, wherein the notch filtering produces a filtered signal at an output of the transformer, the method

further comprising; receiving, by a multiplier connected directly to an output of the transformer, the filtered signal; and multiplying, by the multiplier, an amplitude of the filtered signal.

16. The method of claim 15, wherein the notch filtering comprises: notch filtering the input signal at a second capacitive cross-coupling of the first winding and the second winding.

17. The method of claim 16, wherein the first capacitive cross-coupling includes a first capacitor having a first terminal connected to an input terminal of the first winding and a second terminal connected to an output terminal of the second winding; and the second capacitive cross-coupling includes a second capacitor having a first terminal connected to an output terminal of the first winding and a second terminal connected to an input terminal of the second winding.
